My academic and career objectives lie in advancing trustworthy, efficient, and socially impactful artificial intelligence, particularly in domains where AI intersects with human well-being, healthcare, and mission-critical decision-making. I am driven by the central question: how can we design computationally efficient AI systems that remain reliable, safe, and generalizable in real-world environments? My research interests span Efficient Deep Learning (NLP), Software Engineering and Reliability, Human-centered Interaction, AI in Healthcare, and AI Ethics. To date, I have 5 publications in Q1 journals and IEEE conferences serving as the first author in several of these domains. Alongside my research, I serve as a lecturer in Computer Science at BRAC University, where teaching further strengthens my commitment to academic life.

My long-term goal is to build a career in academia as a professor and researcher contributing meaningfully to scientific advancement while collaborating closely with industry. Over the past few years, my research involvement, passion for teaching, pursuit of deeper scientific questions has strengthened my desire to pursue a Ph.D. in Computer Science. I am applying to the University at Buffalo's CS PhD program to deepen my expertise, engage in impactful collaborative research and continue exploring the questions that drive my work.

Interdisciplinary Research Background: As large language models (LLMs) grew in size and capability, the possibility emerged of developing smaller models that perform comparably to, or even better than, their larger counterparts. Motivated by this idea, I initiated the research project [2], related to creating a compact LLM that could retain strong performance across language tasks. To achieve this, I adopted the innovative framework proposed by Nobel Laureate Dr. Hinton, known as the Knowledge Distillation (KD) method. More specifically, I implemented a **Self-Supervised Teacher-Student KD** approach, where a student model learns to mimic the teacher model using the WikiText-2 dataset. Under the supervision of Prof. Alam, I developed the **LastBERT** model with only 29M parameters. The model is 73.64% smaller compared to the teacher model (BERT-Large-340M), achieving 86-90% of the teacher model's performance. I evaluated the model on GLUE (General Language Understanding Evaluation) datasets. LastBERT excelled in tasks such as paraphrase identification, sentiment analysis, text classification and question answering. In the second phase of this work [2], I applied LastBERT in a real-world use case, involving mental health. The model achieved 85% accuracy in detecting ADHD severity related concerns, classifying either Mild or Severe on the Reddit Mental Health dataset. This is comparable to ClinicalBERT-120M (87%) and DistilBERT-67M (86%). I acted as the first author and corresponding author of this paper and published it in the PLOS ONE Journal (Q1, IF:2.9, CiteScore: 5.6, SJR: 0.803). This project showcases my expertise in **Efficient Deep Learning** (Model Compression), **Natural Language Processing**, and **AI in Healthcare**.

Another of my major research projects [1] demonstrates my ability to lead as the first author in the domains of **Computer Vision**, **Bioimage Informatics**, **Computational Entomology** and **Open-Set Recognition**. Supervised by Prof. Khan, this project addresses a pressing public health challenge in Bangladesh, where vector-borne diseases such as dengue and malaria remain active throughout the year due to the prevalence of specific mosquito species. The goal of this project was to develop an AI agent capable of classifying different mosquito variants responsible for transmitting these diseases. The first step involved curating a dataset of 5000 images and consisting of classes of Aedes, Anopheles, and Culex (Total 10 species) in collaboration with the Department of Biology at Dhaka University and various public image repositories. Using this dataset, 7 models including ViT-base-16, MobileViT, Swin-B, CvT-13, Swin-S, Xception and ResNet50 were fine-tuned for classification. These models achieved highly promising accuracies ranging from 94% to 99%. However, a **key challenge emerged: the closed-set learning problem**, where models incorrectly classified out-of-distribution insects such as flies and bees as mosquitoes. To address this issue, I applied a novel open-set learning approach, a modified OpenMax algorithm. This resulted in a maximum of

89% accuracy while detecting outliers as unknown. This paper was published in PLOS Computational Biology Journal (Q1, IF:4.3, CiteScore: 7.2, SJR: 1.503).

Research Project [3] falls in the realm of **Software Engineering** and NLP. Here I created a labeled dataset of software engineers' profiles and fine-tuned LLMs (RoBERTA and DistilBERT) on this dataset to support automated personnel selection. This work was conducted in collaboration with industry expert, Dr Zia (SINTEF Digital, Oslo, Norway) under the supervision of Prof. Alam. This paper has been published in IEEE Xplore (ICCIT-2024). The extended version of this work [6] involves a hybrid approach of Fuzzy-TOPSIS and LLMs which is currently under review in IEEE ACCESS Journal (Q1).

Finally, papers [4] and [5] reflect my independent exploration of **Clinical NLP** and NLP for **Social Computing**. In Paper [4], I addressed the challenge of multi-class disease classification by leveraging advanced language models such as XLNet and BioBERT to categorize Neoplasms, Cardiovascular, Nervous System, and Digestive Disorders from medical text. This is critical for automated clinical decision support. In paper [5], I investigated the growing problem of misinformation by developing an approach that compares traditional machine learning (SVM with advanced text vectorization) against transformer-based models like BERT for detecting false information propagation. Together, these works highlight my ability to identify socially relevant problems and apply NLP techniques to develop impactful, domain-specific solutions.

Research Vision: Across my work, I have developed a strong interest in creating AI systems that are both computationally efficient and resilient when exposed to real-world uncertainty. During my Ph.D., I hope to deepen this direction by exploring three core areas: (i) efficient and trustworthy deep learning methods, (ii) agentic AI systems that can make contributions in software engineering, and (iii) multimodal and parameter-efficient learning approaches that enable foundation models to adapt reliably across diverse tasks while using limited data and computational resources. These research areas are becoming increasingly important for deploying AI in high impact domains such as healthcare, software engineering, safety-critical operations, and human-centered decision support. My goal is to contribute to the development of reliable, generalizable, and practically deployable AI models that can operate responsibly in complex real-world environments.

Motivation for Pursuing a PhD: My long-term goal is to build an academic career where I can lead impactful research and mentor future scholars, just as my professors have inspired me. Having worked as a lecturer for the past two years, I have developed a strong passion for teaching and research. Pursuing a Ph.D. will provide me with the advanced training, structured research environment, and focused mentorship necessary to deepen my expertise, address meaningful problems, and contribute more robustly to my field.

Research fit with University at Buffalo: University at Buffalo's distinguished CS Ph.D. program, its strong academic structure, vibrant community of students, and prominent faculty members have motivated me to pursue my graduate studies here. I would be honored to work with any faculty member who considers me a good fit for their research group. I am particularly drawn to **Dr. Jinjun Xiong's** *research in Novel AI Algorithms on Computer Vision, which aligns with my interest in efficient AI methods for complex scientific problems*. I am also very interested in the work of **Dr. Harrison Goldstein** on *Software Engineering and Human-Computer Interaction, which aligns with my goal of developing AI-assisted tools for bug detection, program analysis, and improving the reliability of software systems*. In addition, I find the work of **Dr. Haipeng Cai** on Software Engineering integrating *Deep Learning and NLP techniques, and System Security is highly relevant to my experience [3] & [6]*. I believe that my research background aligns well with these faculty members' work, and my problem-solving abilities and eagerness to learn would allow me to contribute meaningfully to their ongoing and future research efforts.

In conclusion, I believe pursuing a PhD at UB is the next step in my academic journey. It is the ideal platform for me to further my impact as a researcher and educator. My prior academic excellence and teaching experience at two of Bangladesh's top universities underscore my commitment to learning and teaching. With a strong foundation of research experience, including publications in Q1 journals, IEEE conferences and innovative contributions to AI, I bring the skills, determination, and drive required to excel in a doctoral program. I am confident that my expertise, combined with UB's exceptional reach, environment, resources, and faculty, will enable me to make meaningful contributions to the field of CS and beyond. I am ready to embrace the challenges and opportunities of doctoral study, and I look forward to contributing to the research community at the University at Buffalo and growing as a researcher within its collaborative environment.

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